$$\mathcal{L}\left[5\ddot{x}(t) + \frac{1}{C}\dot{x}(t) + 2x(t)\right] = \mathcal{L}\left[A \cdot \sin(Bt)\right]$$

$A = 6 \qquad x(0) = 1 = x_0$

$B = 5 \qquad \dot{x}(0) = 2 = x_1$

$C = 11$

$$5\bar{x}s^2 - 5x_0 s - 5x_1 - \frac{1}{11}s\bar{x} - \frac{1}{11}x_0 + 2\bar{x} = \frac{6}{s} + \frac{5}{s^2+25}$$

$$5\bar{x}s^2 - 5s - 10 - \frac{1}{11}s\bar{x} - \frac{1}{11} + 2\bar{x} = \frac{6}{s} + \frac{5}{s^2+25}$$

$$\bar{x}\left(5s^2 - \frac{1}{11}s + 2\right) = \frac{6}{s} + \frac{5}{s^2+25} + 5s + 10 + \frac{1}{11}$$

$$\bar{x} = \frac{55s^4 + 111s^3 + 1441s^2 + 2830s + 1650}{11s\,(s^2+25)\left(5s^2 + \frac{s}{11} + 2\right)} = \frac{A}{11s} + \frac{Bs+C}{s^2+25} + \frac{Ds+E}{5s^2 + \frac{s}{11} + 2}$$

s^4: $55A + 605B + 121D = 55$

s^3: $A + 11B + 605C + 121E = 111$

s^2: $197A + 242B + 11C + 3025D = 1441$

s^1: $25A + 242C + 3025E = 2830$

$550A = 1650$

Po rozwiązaniu systemu równań metodą Kramera:

$$\begin{cases} A = 3 \\ B = \dfrac{-5}{1830634} \approx -2{,}73 \cdot 10^{-6} \\ C = \dfrac{-6765}{1830634} \approx -0{,}0037 \\ D = \dfrac{-18306065}{20136974} \approx -0{,}91 \\ E = \dfrac{100\,900\,676}{110\,753\,357} \approx 0{,}91 \end{cases}$$

$$L^{-1}[\bar{X}] = L^{-1}\left[\frac{3}{11s} - \frac{2{,}73 \cdot 10^{-6} \cdot s + 0{,}0037}{s^2 + 25} + \frac{-0{,}91s + 0{,}91}{5s^2 + \frac{s}{11} + 2}\right]$$

$$x = \frac{3}{11} - 2{,}73 \cdot 10^{-6} \cos 5t - 0{,}00074 \sin 5t - 0{,}0292636\, e^{-\frac{t}{22}} \cdot$$

$$\cdot \left(\sqrt{967} \cdot \cos \frac{\sqrt{967}}{22} t - 0{,}36 \sin \frac{\sqrt{967}}{22} t\right)$$